Calculating Descriptive Statistics

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Table of contents

Objectives . . . 1
Measures of Central Tendency . . . 2
The Mean . . . 2
The Weighted Mean . . . 3
The Median . . . 4
The Mode . . . 5
The Shapes of Distributions . . . 6
Approximately symmetric: the US unemployment rate . . . 6
Left-skewed: graduate student GPA . . . 8
Right-skewed: Nevada adjusted gross income . . . 9
Which measure should you use? . . . 10
Doing this in Excel . . . 10
Measures of Variability . . . 11
The Range . . . 11
The Variance and the Standard Deviation . . . 12
Using the Mean and Standard Deviation Together . . . 12
The Coefficient of Variation . . . 12
The z-score . . . 14
The Empirical Rule . . . 15
Chebyshev's Theorem . . . 16
Working with Grouped Data . . . 18
Application: adjusted gross income in zip code 89015 . . . 18
Measures of Relative Position . . . 19
Percentile rank . . . 20
Quartiles and the interquartile range . . . 20
Box-and-whisker plots . . . 21
The five-number summary . . . 22
Measures of Association Between Two Variables . . . 22
Laboratory 3 . . . 24
Path 1 — upload the files . . . 24
Path 2 — let Claude fetch the files . . . 24

Objectives

- Calculate the mean, weighted mean, median, and mode.
- Choose the measure of central tendency that describes the data honestly.
- Calculate and interpret the range, variance, and standard deviation.
- Use the coefficient of variation, the z-score, the empirical rule, and Chebyshev's Theorem.
- Calculate the mean and standard deviation of grouped data.

- Calculate percentiles, percentile rank, and quartiles.
- Construct and interpret box-and-whisker plots.
- Calculate and interpret the covariance and the correlation coefficient.

> Chapter 2 asked you to read a picture. This chapter asks you to read a *number*, which is harder, because a number carries no visible warning that it is the wrong number for the question.
>
> The formulas below stay in the body of the note rather than being folded away. You are not assessed on writing the code — the code that draws the histogram is folded, and you may ignore it. You are assessed on knowing what a standard deviation is, and on saying which of two numbers a decision should rest on.

Measures of Central Tendency

Central tendency is a single value used to describe a central focal point in a data set.

The Mean

Given a data series $X = \{x_1, x_2, ..., x_n\}$, the sample **mean** is

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i,$$

where n is the number of members of the sample. For a population we write

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i,$$

where N is the number of members of the population.

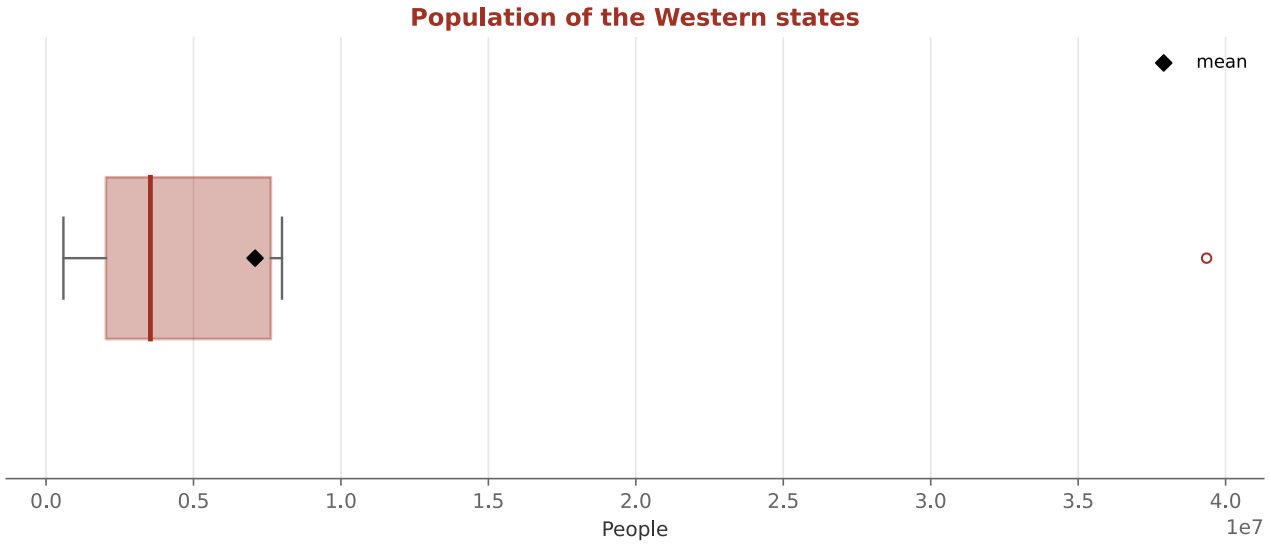
The mean is easy to compute, easy to explain, and always produces exactly one number. That is its entire case for the defence. Here is the case against.

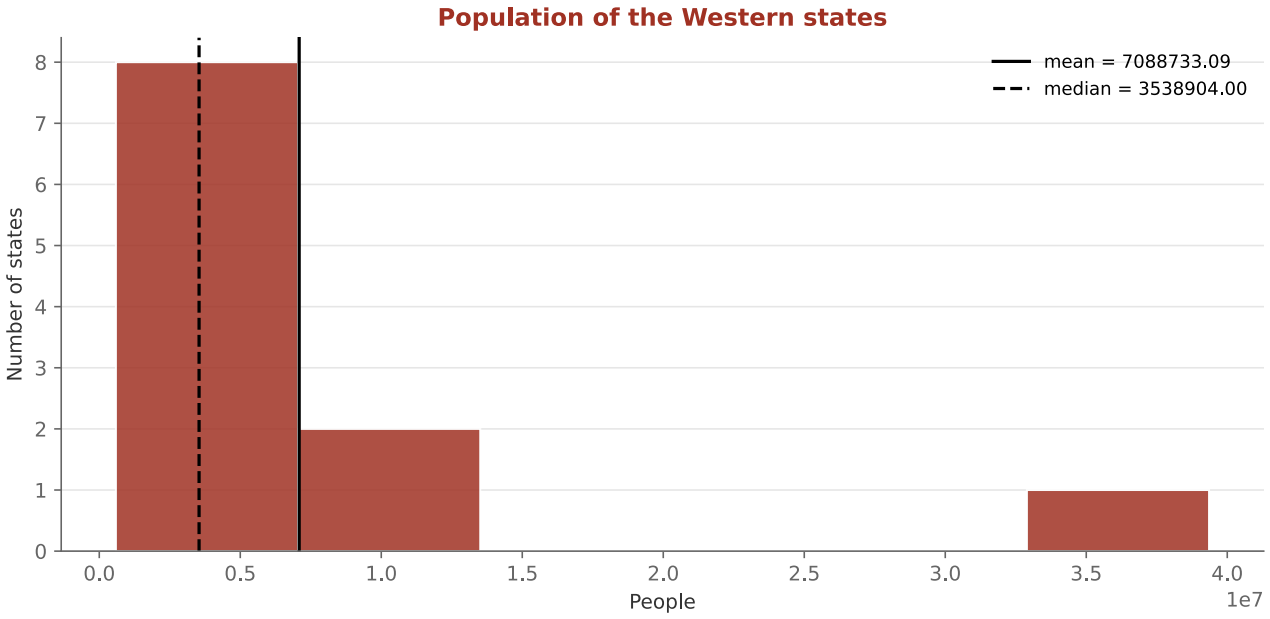
Example: the population of the Western states

	Member	People
0	WY	588,753
1	MT	1,144,694
2	ID	2,029,733
3	NM	2,125,498
4	NV	3,282,188
5	UT	3,538,904
6	OR	4,273,586
7	CO	6,012,561
8	AZ	7,623,818
9	WA	8,001,020
10	CA	39,355,309

The mean population of a Western state is 7,088,733, as of 2025-01-01. But only 3 of the 11 Western states have a population larger than that mean, and 8 have a smaller one.

> A "central" value that 8 of 11 observations fall below is not describing a centre. It is describing California.





> **Outliers** are values much higher or lower than most of the data. The mean is pulled toward them. The median is not.

The Weighted Mean

> For a data set $\{x_1, x_2, ..., x_n\}$ with corresponding weights $\{w_1, w_2, ..., w_n\}$, the **weighted mean** is
>
> $$\bar{x}_w = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i}.$$

This is the formula MBAs get wrong most often in practice, usually by not reaching for it at all. Consider the unemployment rate of the fifty states.

	measure	unemployment rate (%)
0	Unweighted mean of the 50 state rates	3.97
1	Population-weighted mean	4.32

The two answers differ by +0.35 percentage points, and only one of them is the unemployment rate of the country. The unweighted mean gives WY — population 588,753 — exactly the same say as CA, population 39,355,309.

> Whenever the things you are averaging represent groups of different sizes — states, stores, regions, customer segments — the plain average answers the question *"what is the typical store's rate?"* and the weighted average answers *"what is the rate across all customers?"* Those are different questions. Decide which one you were asked before you pick a formula.

The Median

> The **median** is the value for which half the observations are higher and half are lower.
>
> The **index point** is the position of the median once the data are arranged in ascending order:
>
> $$i = 0.5\,n,$$
>
> where n is the number of observations. If i is not an integer, round up and take the value in that position. If i is an integer, take the average of the values in positions i and $i + 1$.

The Western states again

The median population of a Western state is 3,538,904. The mean is 7,088,733 — higher by 3,549,829.

The list is already sorted above. With $n = 11$, the index point is \$0.5 \times 11 = 5\.5\$, which is not an integer, so we round up to position 6 and read off the value there.

An even number of observations

New England has six states, so the index point lands exactly on an integer and the second half of the rule applies.

	Member	People	position
0	VT	644,663	1
1	RI	1,114,521	2
2	ME	1,414,874	3
3	NH	1,415,342	4
4	CT	3,688,496	5
5	MA	7,154,084	6

With $n = 6$, the index point is $0.5 \times 6 = 3$. Three is an integer, so the median is the average of the third and fourth values:

$$\frac{1}{2}(1,414,874 + 1,415,342) = 1,415,108$$

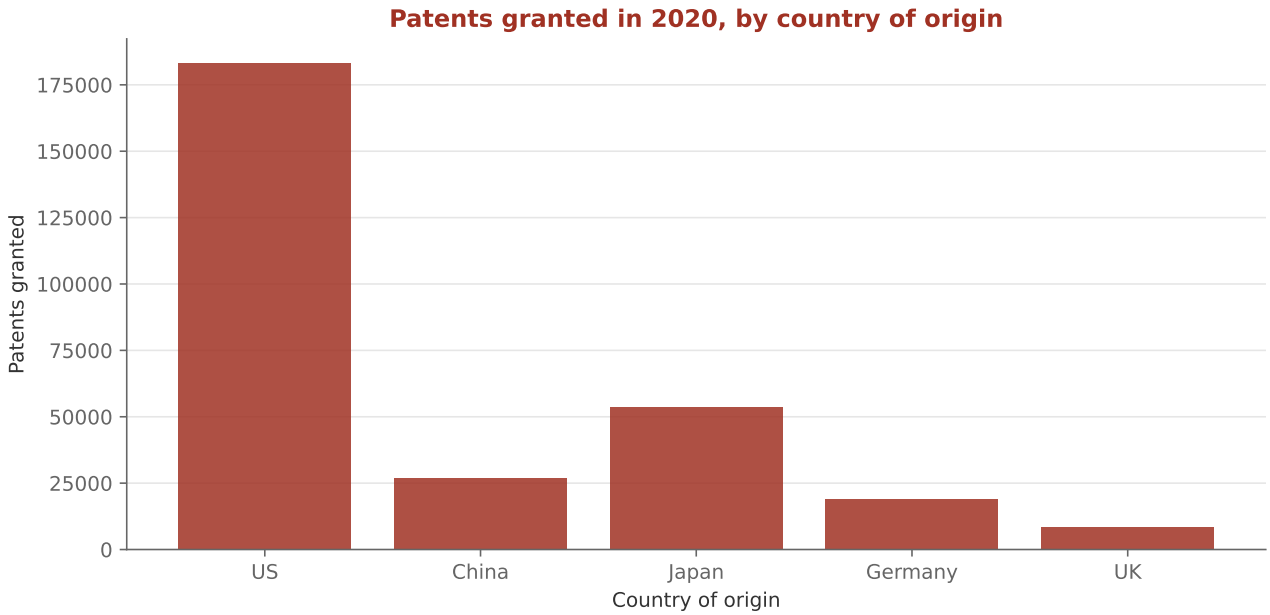
The mean is 2,571,997 and the median is 1,415,108, a difference of 1,156,889. Compare that to the 3,549,829 gap out West. Same statistic, same country, a far smaller disagreement — because New England has no California in it.

The Mode

> The **mode** is the value that appears most often in a data set. It is the only measure of central tendency that works on categories.

Example: US patents granted, by country of origin

year	US	China	Japan	Germany	UK
2009	95,032	2,270	38,066	10,353	4,011
2010	121,164	3,303	46,978	13,633	5,038
2011	121,247	3,786	48,256	12,968	4,924
2012	134,187	5,341	52,773	15,041	5,876
2013	147,652	6,597	54,170	16,605	6,551
2014	158,709	7,921	56,006	17,594	7,156
2015	155,982	9,004	54,422	17,752	7,167
2016	160,540	11,737	51,938	17,307	7,345
2017	169,208	14,853	51,952	18,190	7,686
2018	161,809	16,759	49,702	17,326	7,599
2019	186,118	22,962	55,927	19,850	9,041
2020	183,147	26,845	53,779	19,173	8,470



The modal country of origin is US, with 183,147 patents — 62.8% of the five countries shown.

This series runs from 1992 through 2020. It is not current, and the FRED page it comes from does not put a banner on the chart telling you so. Chapter 2 asked you to check the last observation date of every series you use. This is the series that punishes you for skipping that step.

Notice also that a mean of these five numbers would be a nonsense statistic — the average of the US, China, Japan, Germany and the UK is not a country. When your variable is a category, the mode is the only measure of central tendency available to you.

The Shapes of Distributions

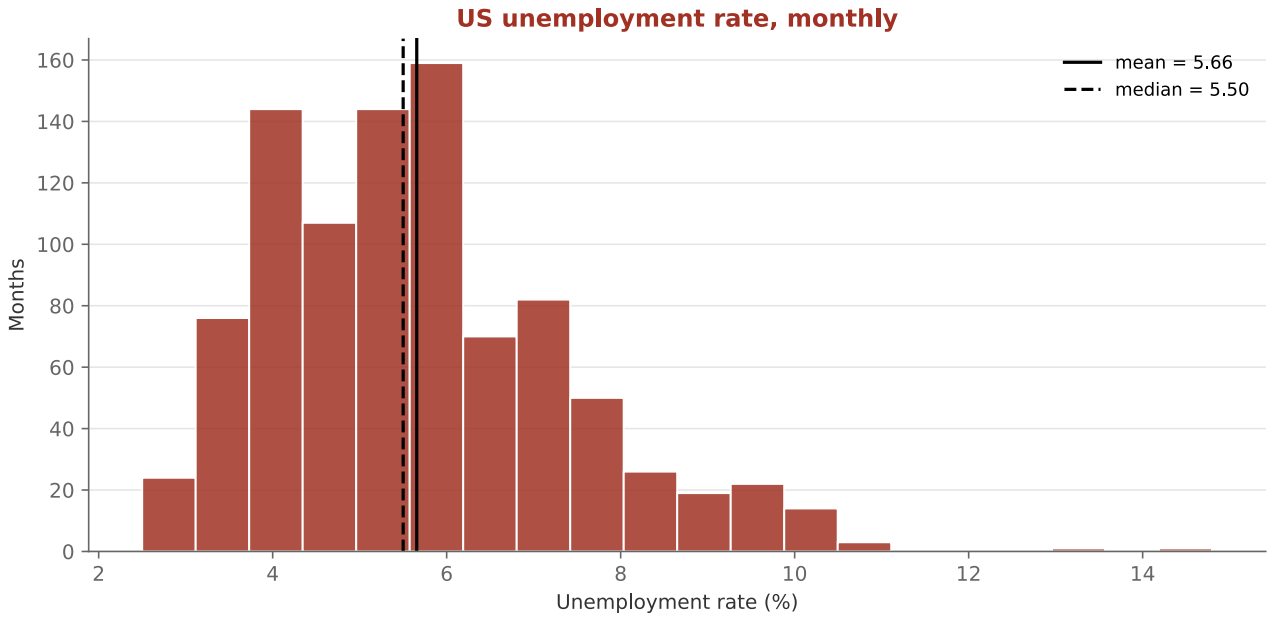
A **symmetrical** distribution is a mirror image about its centre. Its mean and median are approximately equal.

In a **left-skewed** distribution, the median is larger than the mean.

In a **right-skewed** distribution, the median is smaller than the mean.

The single most useful diagnostic in this chapter is free: compute the mean and the median and compare them. If they are far apart, one of them is lying to you about the middle.

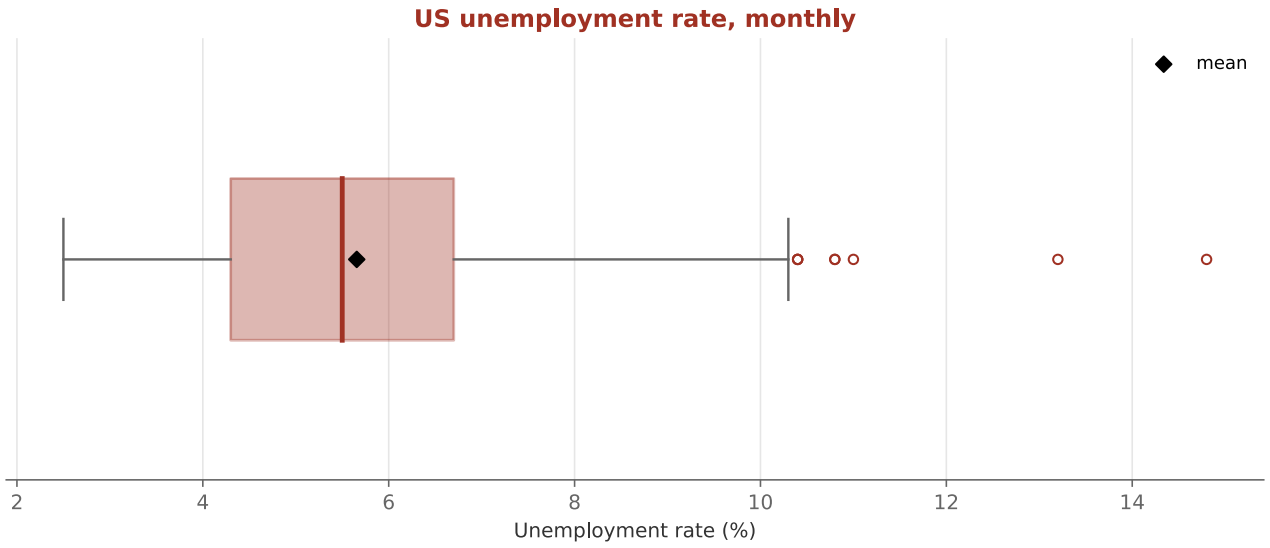
Approximately symmetric: the US unemployment rate



	UNRATE
n	942.0000
mean	5.6562
median	5.5000
mode	5.4000
min	2.5000
max	14.8000
range	12.3000

	UNRATE
variance	2.9034
std_dev	1.7039
CV_percent	30.1252

Watch the `mode` row across this chapter. It carries a value here only because FRED publishes the unemployment rate rounded to one decimal place, so months tie by accident of rounding rather than because the economy has a favourite number. On an unrounded measurement — the return series later in this chapter — the same row reads `NaN`, because no value repeats and there is no mode to report. That blank is the honest answer. The mode earns its keep on categories, as it did with the patent data above, and rarely anywhere else.



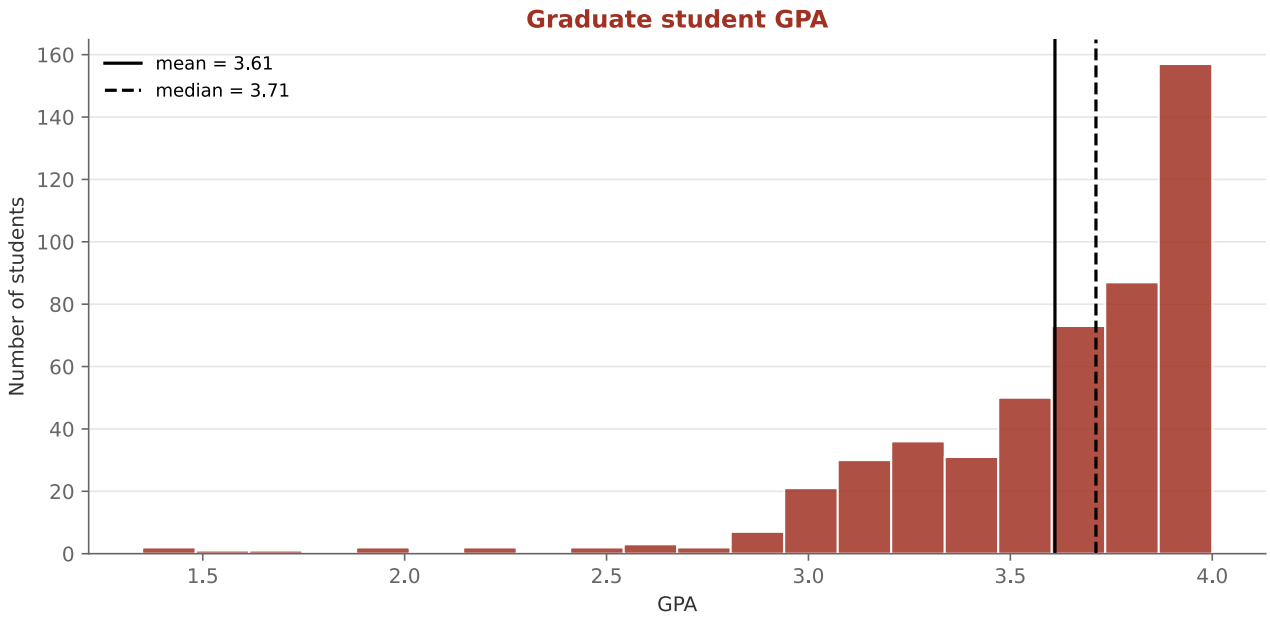

```
US unemployment rate: mean 5.656, median 5.5, difference +0.1562 -- right-skewed.
Coverage: January 1948 to July 2026  (942 months)
```

The mean is 5.66 and the median is 5.50 — a gap of +0.16 percentage points, which is 0.09 of a standard deviation. That is what approximate symmetry looks like in numbers rather than in a picture: not identical, but close enough that either one would serve as “the typical month.”

In this series the mean sits slightly above the median, which makes the right tail marginally the longer one — but a gap that small is not a finding, and it can change sign as more months are added.

What is not in doubt is the mechanism. Unemployment rises hard and fast in a recession and comes back down slowly, so the months far from the centre tend to be the high ones, and the box plot shows those sitting past the upper fence. Compare that to the GPA and income distributions below, where the gap between mean and median is large enough to survive any amount of new data.

Left-skewed: graduate student GPA



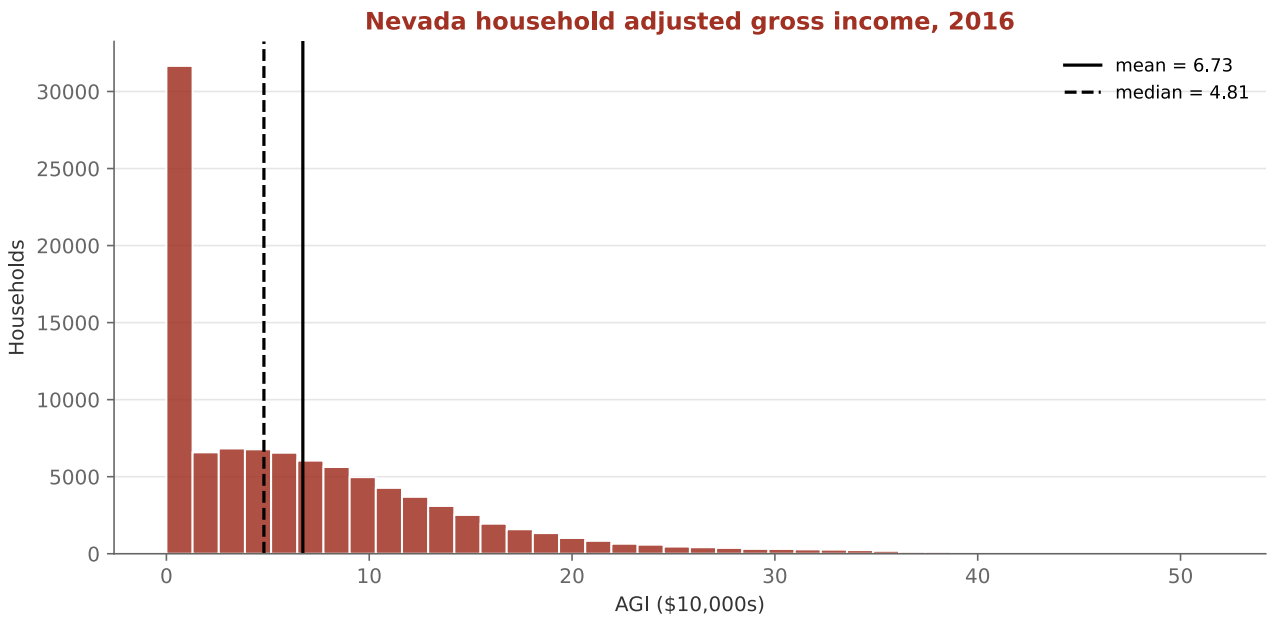

```
Graduate GPA: mean 3.61, median 3.712, difference -0.1015 -- left-skewed.
```

	GPA
n	507.0000
mean	3.6105
median	3.7120
mode	4.0000
min	1.3500
max	4.0000
range	2.6500
variance	0.1612
std_dev	0.4015
CV_percent	11.1215

The median exceeds the mean, which is the definition of left skew. The cause is visible in the histogram: GPA has a hard ceiling at 4.0 and no floor to speak of, so the only direction an unusual student can be unusual *in* is downward. A few very low GPAs drag the mean below the median while the bulk of the distribution piles up against the ceiling.

> Any measure with a ceiling tends to skew left: exam scores, on-time delivery rates, satisfaction surveys capped at 5. Any measure with a floor at zero and no ceiling tends to skew right: income, insurance claims, time on hold, company size. Knowing which one you are holding tells you what to expect before you compute anything.

Right-skewed: Nevada adjusted gross income




```
Nevada AGI ($10,000s): mean 6.727, median 4.808, difference +1.919 -- right-skewed.
n = 100,000 households in this sample
```

	AGI ($10,000s)
n	100000.0000
mean	6.7271
median	4.8076
mode	0.0000
min	0.0000
max	51.6211
range	51.6211
variance	55.1541
std_dev	7.4266
CV_percent	110.3987

The mean household AGI is $67,271 and the median is $48,076. Only 40.3% of households earn more than the *average* household.

This is the shape of nearly every income distribution ever measured, and it is why “average income” is a number to distrust in a policy argument. If you are building a case about typical households, the median is the honest statistic and you should say which one you used.

Two notes on this data set, both of which you would want stated if you were the reader rather than the writer.

First, these observations are **simulated** from the counts the IRS publishes by zip code — the IRS does not release household-level returns. The shape is real; no individual row is a real household.

Second, this file is a random sample of 100,000 households drawn from roughly 1.3 million. Sampling is Chapter 7's subject, and the reason a sample this size is safe here is exactly what that chapter is about.

Which measure should you use?

	Advantage	Disadvantage
Mean	Easy to compute and to explain; uses every observation	Pulled by outliers
Median	Unaffected by outliers	Ignores the size of the values on either side
Mode	The only option for categorical data	May not exist, and may not be unique

The practical rule: report both the mean and the median. When they agree, you have lost nothing. When they disagree, you have found the most important sentence in your write-up.

Doing this in Excel

Many of you will present this work in Excel, and the formulas are direct:

Statistic	Excel
Mean	`=AVERAGE(range)`
Median	`=MEDIAN(range)`
Mode	`=MODE.SNGL(range)`
Weighted mean	`=SUMPRODUCT(values, weights)/SUM(weights)`
Standard deviation (sample)	`=STDEV.S(range)`
Variance (sample)	`=VAR.S(range)`
Percentile	`=PERCENTILE.EXC(range, k)`
Percentile rank	`=PERCENTRANK.EXC(range, value)`
Quartile	`=QUARTILE.EXC(range, quart)`
Correlation	`=CORREL(range1, range2)`

Excel's **Data Analysis** add-in produces the whole descriptive block at once: **Data → Data Analysis → Descriptive Statistics**, tick *Summary statistics*. If the Data Analysis button is not on your ribbon, it is an add-in you enable under **File → Options → Add-ins → Analysis ToolPak**.

You do not have to build the workbook by hand:

PROMPT

> Using `data/grad_student_gpa.csv` from https://github.com/jcrooker/mba775, build an Excel workbook with two sheets. Sheet 1 holds the raw GPA data. Sheet 2 computes the mean, median, mode, sample standard deviation, sample variance, minimum, maximum, range, and the first and third quartiles — each with a real Excel formula referencing Sheet 1, not a pasted value, so I can change the data and watch the summary update. Label every row. Give me the file to download.

`PERCENTILE.EXC` and `PERCENTILE.INC` are different functions and give different answers. This course uses the `.EXC` form, which matches the index point rule defined below. If your number disagrees with a classmate's in the

fourth decimal place, this is usually why — and being able to say *which* rule produced your number is the actual skill.

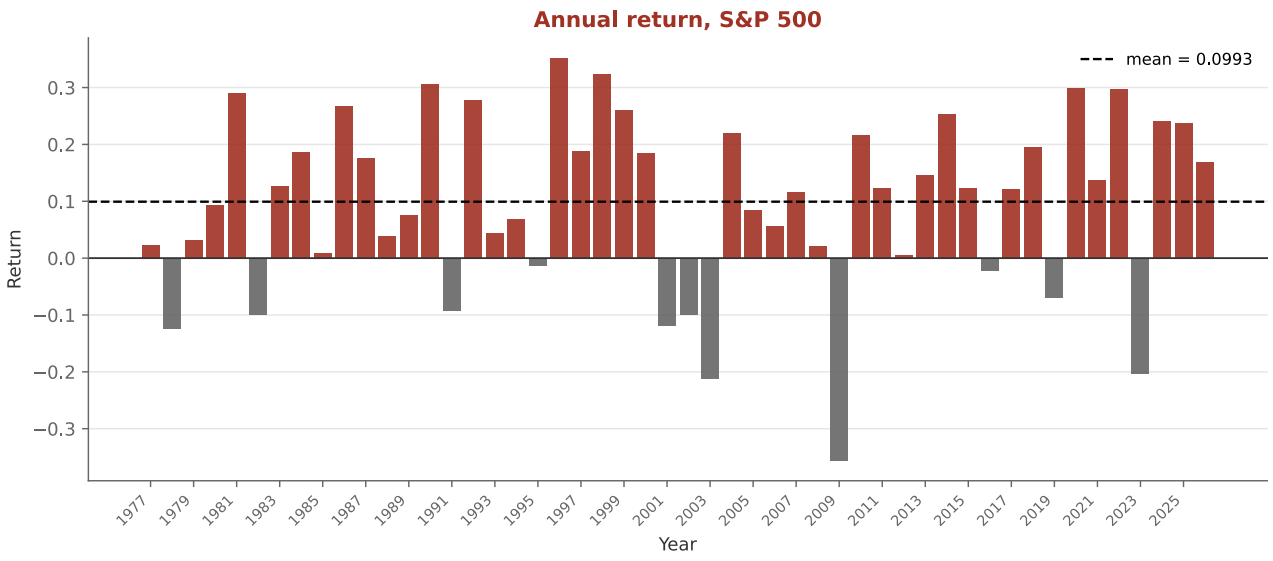
Measures of Variability

Two data sets can share a mean and have nothing else in common. Variability is what separates them.

The Range

The **range** is the largest value minus the smallest value.

Example: annual returns to the S&P 500



	S&P 500 annual return
n	50.0000
mean	0.0993
median	0.1222
mode	NaN
min	−0.3561
max	0.3520
range	0.7081
variance	0.0246
std_dev	0.1568
CV_percent	157.9911

Over these 50 years the range is 0.7081 — 70.8 percentage points separating the best year (1996, at +35.2%) from the worst (2009, at −35.6%).

The range is built from exactly two observations and ignores the other 48. Add one catastrophic year and the range moves; add fifty ordinary years and it does not move at all. It is a useful headline and a poor summary.

The Variance and the Standard Deviation

For a *sample*, the **variance** is

$$\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

The **standard deviation** is its square root:

$$\hat{\sigma} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}.$$

The standard deviation is carried in the same units as the original data — the variance is in units *squared*, which is why nobody reports "the variance in returns was 0.0246 return-squared." Square it back and report the standard deviation.

The divisor is $n-1$, not n. The sample variance computed this way is an **unbiased estimator** of the population variance: repeat the sampling many times and the average of the sample variances converges on the true value. Divide by n and you will, on average, understate the spread.

In Excel this is the difference between `STDEV.S` and `STDEV.P`. In pandas it is the `ddof` argument, and pandas defaults to the sample version while numpy defaults to the population version. Two libraries in the same language, two different answers. Know which one you called.

	divisor	standard deviation
0	n - 1 (sample, Excel STDEV.S)	0.156811
1	n (population, STDEV.P)	0.155235

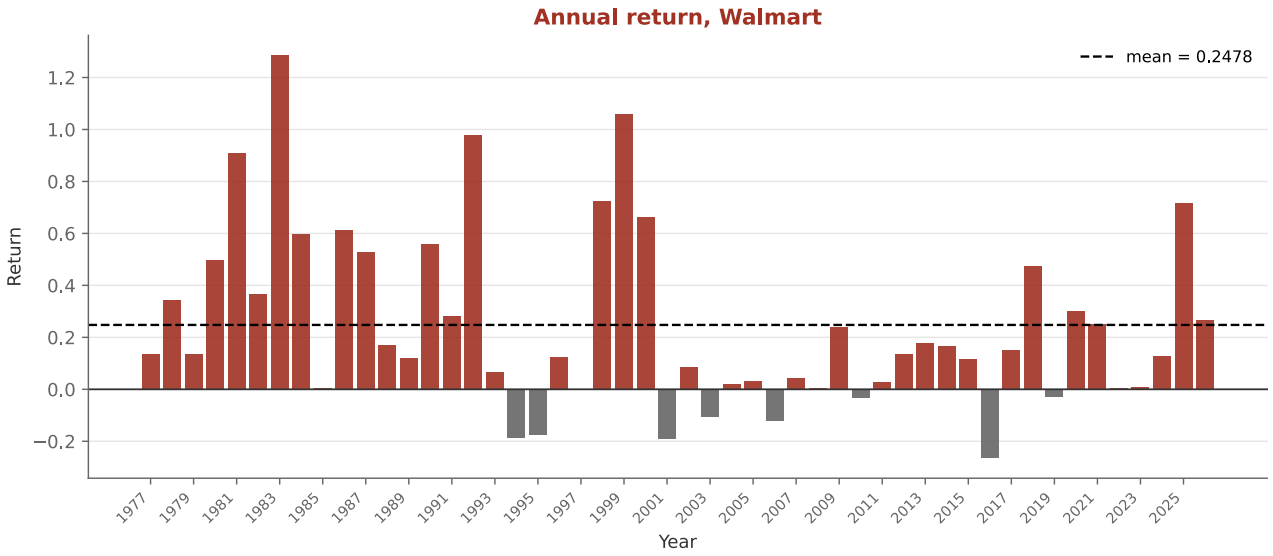
Using the Mean and Standard Deviation Together

The Coefficient of Variation

The **coefficient of variation** measures the standard deviation as a percentage of the mean:

$$CV = \frac{\hat{\sigma}}{\bar{x}} \cdot 100.$$

Example: the S&P 500 against Walmart



	S&P 500	Walmart
n	50.0000	50.0000
mean	0.0993	0.2478
median	0.1222	0.1358
std_dev	0.1568	0.3443
range	0.7081	1.5496
CV_percent	157.9911	138.9323

Written out for each:

$$CV_{SP} = \frac{0.1568}{0.0993} \cdot 100 = 158.0$$

$$CV_{WMT} = \frac{0.3443}{0.2478} \cdot 100 = 138.9$$

Read those two lines together.

Walmart's standard deviation is 2.20 times the S&P 500's, so by standard deviation alone Walmart is the more volatile holding — and that is true. But Walmart's *coefficient of variation* is 138.9 against the index's 158.0, because the mean return it is deviating around is 2.5 times larger. Per unit of return earned, the S&P 500 carried the more variable ride.

An MBA who takes only the standard deviation away from this chapter will rank those two backwards. The standard deviation answers *"how much does it move?"* The coefficient of variation answers *"how much does it move, given what it pays?"* — and the second question is the one an investment committee is actually asking.

The coefficient of variation is only interpretable when the mean is positive and comfortably away from zero. As $\bar{x}$ approaches zero the ratio explodes, and for a mean that can be negative — a spread, a net change, a

margin swing — the CV is meaningless and occasionally sign-flipped. Check the mean before you quote the ratio.

The z-score

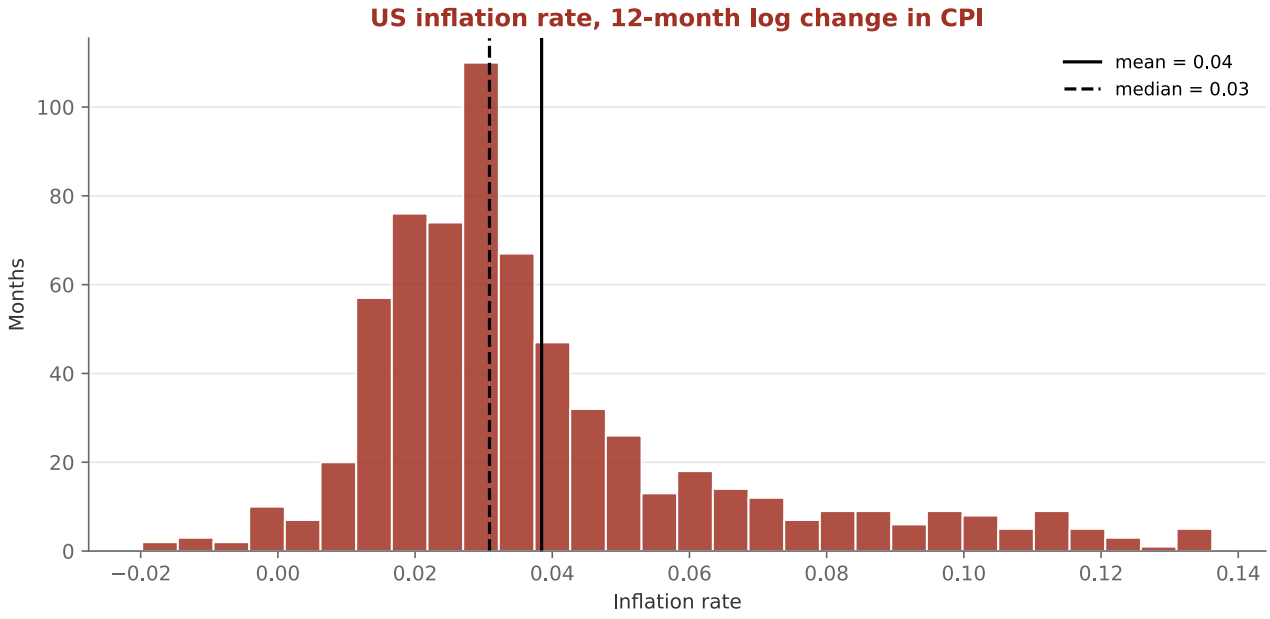
The **z-score** is the number of standard deviations a particular value sits from the mean of its distribution:

$$z = \frac{x - \bar{x}}{\hat{\sigma}}.$$

Values with $|z| > 3$ are conventionally categorised as **outliers**.

Expressing a value as a z-score is what makes two things measured in different units comparable at all. It is also the machinery underneath every inference test in the second half of this course, so it repays attention now.

Example: z-scores and inflation



	12-month inflation
n	666.0000
mean	0.0384
median	0.0308
mode	NaN
min	−0.0198
max	0.1362
range	0.1560
variance	0.0007
std_dev	0.0273
CV_percent	71.0293

The highest 12-month inflation rate in this series is 13.62%, in March 1980. Its z-score is

$$z = \frac{x - \bar{x}}{\hat{\sigma}} = \frac{0.1362 - 0.0384}{0.0273} = 3.5800$$

That month sits 3.58 standard deviations above the average month, and does clear the conventional $|z| > 3$ threshold for an outlier.

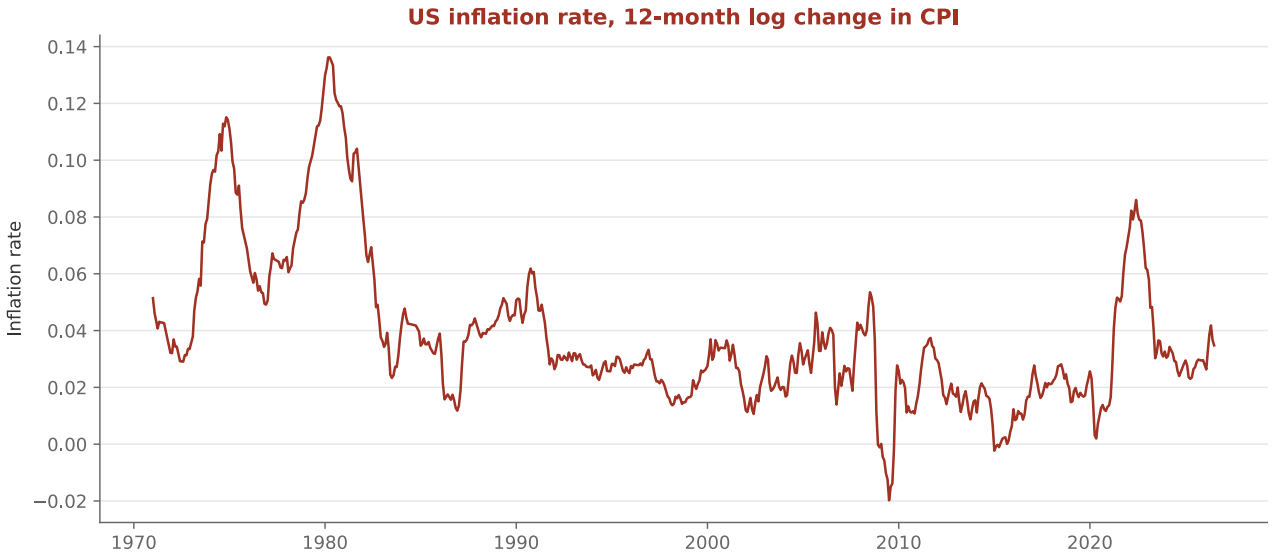
The most recent observation, July 2026, sits at 3.48% — a z-score of −0.13. Two inflation rates, the same units, and the z-score is what lets you say how far apart they really are.

How unusual is unusual?

A z-score is only useful if you can turn it back into a count. Here is every month whose inflation rate was more than two standard deviations above the mean:

```
Threshold (z = 2): 9.31% inflation
Months above it:   47 of 666 (7.06%)

   month  inflation_pct
Mar 1980          13.62
Apr 1980          13.62
May 1980          13.48
Jun 1980          13.34
Feb 1980          13.24
Jan 1980          12.99
Dec 1979          12.45
Jul 1980          12.36
```

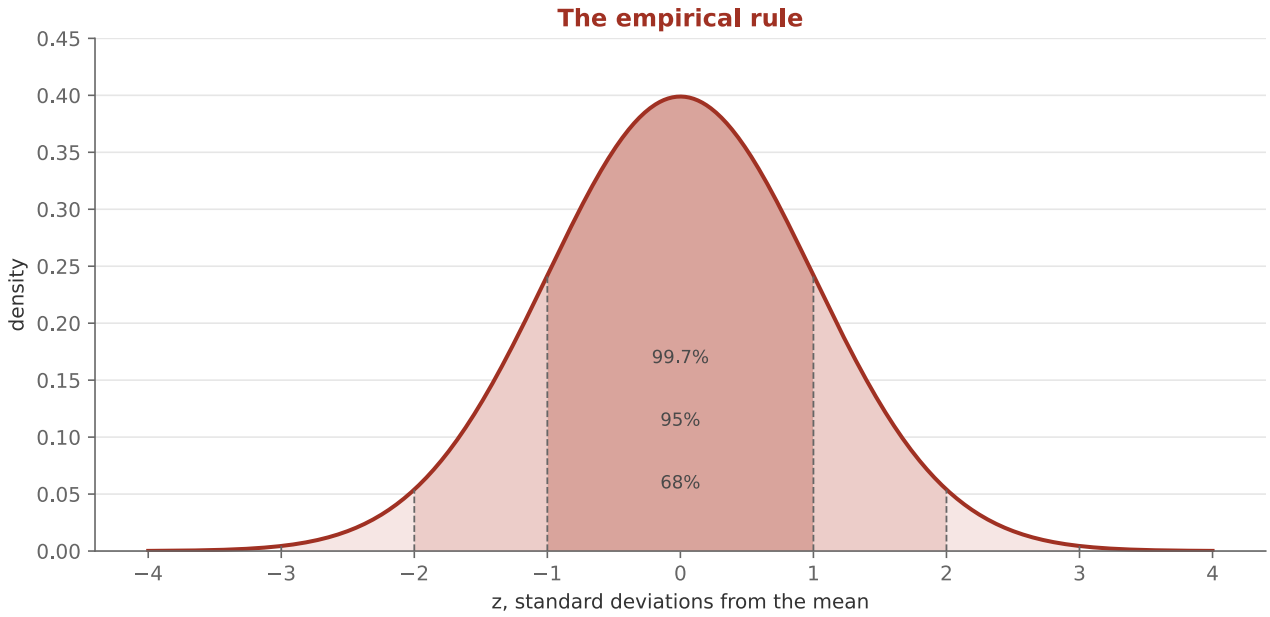


The Empirical Rule

If a distribution is bell-shaped and symmetric, approximately 68%, 95%, and 99.7% of the values fall within one, two, and three standard deviations of the mean, respectively.

Rearranged, a value can be recovered from its z-score:

$$x = \mu + \sigma \cdot z.$$



That picture is of a mathematical function, not of any data set. The rule only applies to the extent your data resembles it, and that is a claim to be checked rather than assumed:

within	interval_low	interval_high	actual_percent	bell_shaped_percent	chebyshev_at_least
1 sd	3.95	7.36	67.62	68.3	0.00
2 sd	2.25	9.06	95.22	95.4	75.00
3 sd	0.54	10.77	99.47	99.7	88.89

within	interval_low	interval_high	actual_percent	bell_shaped_percent	chebyshev_at_least
1 sd	−0.70	14.15	86.03	68.3	0.00
2 sd	−8.13	21.58	95.16	95.4	75.00
3 sd	−15.55	29.01	98.05	99.7	88.89

For the unemployment rate, 67.6% of months fall within one standard deviation of the mean, against the 68.3% the bell curve predicts. For Nevada AGI the figure is 86.0%.

Look at the `interval_low` column of the AGI table. Three standard deviations below the mean is −15.55 — a *negative* income, in a data set where nothing can be below zero. The empirical rule did not fail here; it was never applicable. Applying a symmetric rule to a skewed distribution produces intervals that reach into territory the variable cannot occupy, and nothing in the arithmetic warns you.

Chebyshev's Theorem

Chebyshev's Theorem states that for any $z > 1$, the percentage of values falling within z standard deviations of the mean is at least

$$\left(1 - \frac{1}{z^2}\right) \cdot 100.$$

This is the tool for the situation you are actually in most of the time: a distribution whose shape you do not know. It applies to *every* distribution — skewed, bimodal, lumpy, anything — which is exactly why the guarantee it gives is weak. It is a floor, never an estimate.

Applied to inflation

Take the two standard deviation threshold from the section above. What does Chebyshev guarantee about how often inflation can exceed it?

	Chebyshev at z = 2
z	2.0
at_least_within_percent	75.0
at_most_outside_percent	25.0
at_most_one_tail_percent	12.5

$$\left(1 - \frac{1}{z^2}\right) \cdot 100 = \left(1 - \frac{1}{(2)^2}\right) \cdot 100 = 75.00$$

At least 75.00% of months must lie within two standard deviations of the mean, so at most 25.00% lie outside that interval — and, if we are willing to treat the two tails as comparable, at most 12.50% above it.

In this data, 92.79% of months actually fell inside that interval and 7.06% were above it.

> The bound held, and it held by a wide margin. That is the normal outcome and it is the thing to understand about Chebyshev: it is almost always loose. It tells you what *cannot* be true, not what is likely. A consultant who reports the Chebyshev bound as a forecast has reported a number that is correct and useless at the same time.
>
> Two cautions worth carrying out of here. The theorem bounds both tails together; splitting the remainder in half assumes a symmetry the theorem never promised, so state a one-tail figure as an upper bound and never as an estimate. And for $z \leq 1$ the formula returns zero or worse — it tells you nothing at all, which is honest but unquotable.

Applied to graduate GPA

What fraction of graduate students should we expect to hold a GPA below 3.0? The z-score of a 3.0 is

$$z = \frac{3.0 - 3.6105}{0.4015} = -1.5203$$

	Chebyshev for a 3.0 GPA
z	1.5203
at_least_within_percent	56.7400
at_most_outside_percent	43.2600
at_most_one_tail_percent	21.6300

```
Students below a 3.0 GPA: 24 of 507  (4.73%)
```

Chebyshev bounds the share below 3.0 at no more than 21.63%. The actual figure is 4.73%.

> Note how far apart those two numbers are, and note the direction: the bound is not close, but it is not violated either. That is what a guarantee that holds for *any* distribution costs you. If you are willing to assume a shape, you can say far more — which is what Chapters 5 and 6 are for.

Working with Grouped Data

Sometimes you never see the individual observations. What you are handed is a table of counts by class, which is how most published statistics arrive.

The **midpoint** of a class is the average of its endpoints.

The **sample mean for grouped data** is

$$\bar{x} \approx \frac{1}{n} \sum_{i=1}^{k} f_i \cdot m_i,$$

where f_i is the frequency of class i, m_i its midpoint, $n = \sum_{i=1}^{k} f_i$, and k the number of classes.

The **sample variance for grouped data** is

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^{k} f_i \cdot (m_i - \bar{x})^2.$$

Application: adjusted gross income in zip code 89015

The IRS publishes tax return counts by zip code and income bracket at its SOI zip code data portal. Below are the 2016 counts for zip code 89015 — Henderson, Nevada.

	class	low	high	frequency	midpoint
0	$1 - $25,000	1	25000	6600	12500.5
1	$25,000 - $50,000	25000	50000	4820	37500.0
2	$50,000 - $75,000	50000	75000	2550	62500.0
3	$75,000 - $100,000	75000	100000	1600	87500.0
4	$100,000 - $200,000	100000	200000	1790	150000.0
5	$200,000 - $300,000	200000	300000	310	250000.0

	class	frequency_f	midpoint_m	f_times_m	deviation	f_times_dev_sq
0	$1 - $25,000	6,600	12,500.50	82,503,300.00	−38,921.59	9,998,274,228,656.28
1	$25,000 - $50,000	4,820	37,500.00	180,750,000.00	−13,922.09	934,234,293,508.67
2	$50,000 - $75,000	2,550	62,500.00	159,375,000.00	11,077.91	312,936,326,298.65
3	$75,000 - $100,000	1,600	87,500.00	140,000,000.00	36,077.91	2,082,585,141,918.07
4	$100,000 - $200,000	1,790	150,000.00	268,500,000.00	98,577.91	17,394,512,373,700.81
5	$200,000 - $300,000	310	250,000.00	77,500,000.00	198,577.91	12,224,287,976,509.79
6	TOTAL	17,670		908,628,300.00		42,946,830,340,592.28

Reading the totals off that table:

$$\bar{x} = \frac{1}{17,670}[908, 628, 300] = \$51,422.09$$

$$\hat{\sigma}^2 = \frac{1}{17,669}[42,946,830,340,592] = 2,430,631,633.97$$

$$\hat{\sigma} = \sqrt{2,430,631,633.97} = \$49,301.44$$

> Every observation inside a class has been replaced by that class's midpoint, so this is an approximation and the note should say so. The top class is the worst offender: "$200,000 – $300,000" gets a midpoint of $250,000, and if the published table's top class had been open-ended — "$200,000 or more" — you would have to invent a midpoint, and your standard deviation would inherit whatever you invented.

What share earns above $250,000?

We do not know the shape of this distribution, so Chebyshev is the only tool available.

$$z = \frac{250,000 - 51,422.09}{49,301.44} = 4.0278$$

	Chebyshev at $250,000
z	4.0278
at_least_within_percent	93.8400
at_most_outside_percent	6.1600
at_most_one_tail_percent	3.0800

At least 93.84% of this zip code's returns fall within 4.03 standard deviations of the mean, so no more than 3.08% report AGI above $250,000.

> Check that against the table you started from: the published counts say 310 of 17,670 returns — 1.75% — fall in the top bracket at all. The bound is true and the bound is loose. When you have the counts, count. Chebyshev is for when you do not.

Measures of Relative Position

Where does one value sit relative to the rest?

> A **percentile** measures the approximate percentage of values in the data set that fall below the value of interest.
>
> The **ρth percentile** is the value that at least ρ percent of the observations fall below.
>
> The **index point** for the pth percentile is
>
> $$i = \frac{p}{100} \cdot n.$$
>
> If i is not a whole number, round up; that position holds the value. If i is a whole number, take the midpoint of positions i and $i + 1$.

```
3.925
```

With $n = 507$ students, the index point for the 80th percentile is \$0.80 \times 507 = 405\.6\$. The 80th percentile GPA is 3.9250.

> There is more than one rule for this, and different software uses different ones. Here is the same percentile computed three ways:

	80th percentile
course_rule (Excel PERCENTILE.EXC)	3.925
numpy default (linear)	3.925
nearest rank	3.925

They are close, and on a data set this size the disagreement is in the third decimal. On a data set of twelve observations it would not be. This course uses the index-point rule above, matching Excel's `PERCENTILE.EXC`. The point is not that one is right — it is that you should be able to say which rule produced your number when someone else's differs.

Percentile rank

The **percentile rank** of a value x is

$$\text{Percentile Rank} = \frac{1}{n}\left[\frac{1}{2} + \sum_{i=1}^{n} I\{x_i < x\}\right] \cdot 100,$$

where $I\{\cdot\}$ is an **indicator function** taking the value 1 when the statement inside is true and 0 when it is false.

```
52.96
```

A student with a 3.75 GPA sits at the 52.96th percentile: 268 of 507 students are below them.

Percentile and percentile rank are inverses of one another. The first takes a percentage and returns a value; the second takes a value and returns a percentage. Getting them the wrong way round is the most common error in this section.

Quartiles and the interquartile range

The first, second, and third **quartiles** are the 25th, 50th, and 75th percentiles. The second quartile is the median.

The **interquartile range** is

$$IQR = Q_3 - Q_1,$$

the spread of the middle 50% of the data.

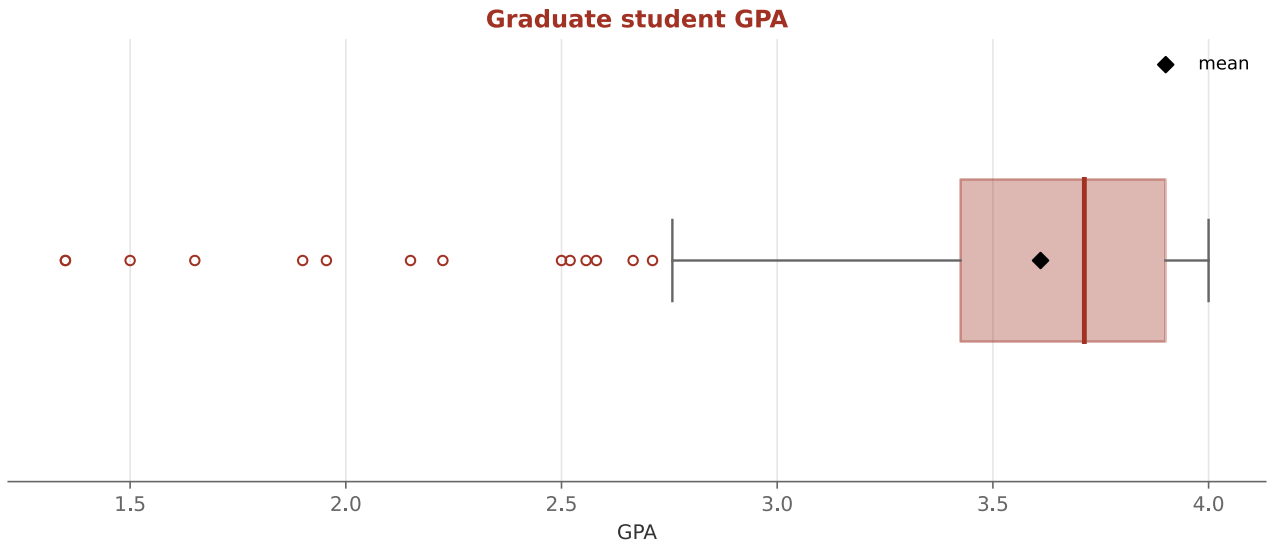
	Graduate GPA
Q1 (25th percentile)	3.4250
Q2 (median)	3.7120
Q3 (75th percentile)	3.9000

	Graduate GPA
IQR	0.4750
lower fence Q1 - 1.5*IQR	2.7125
upper fence Q3 + 1.5*IQR	4.6125

The IQR is a measure of spread with the same virtue the median has: the two most extreme observations in the data set cannot move it. Where the range is built entirely from outliers, the IQR is immune to them.

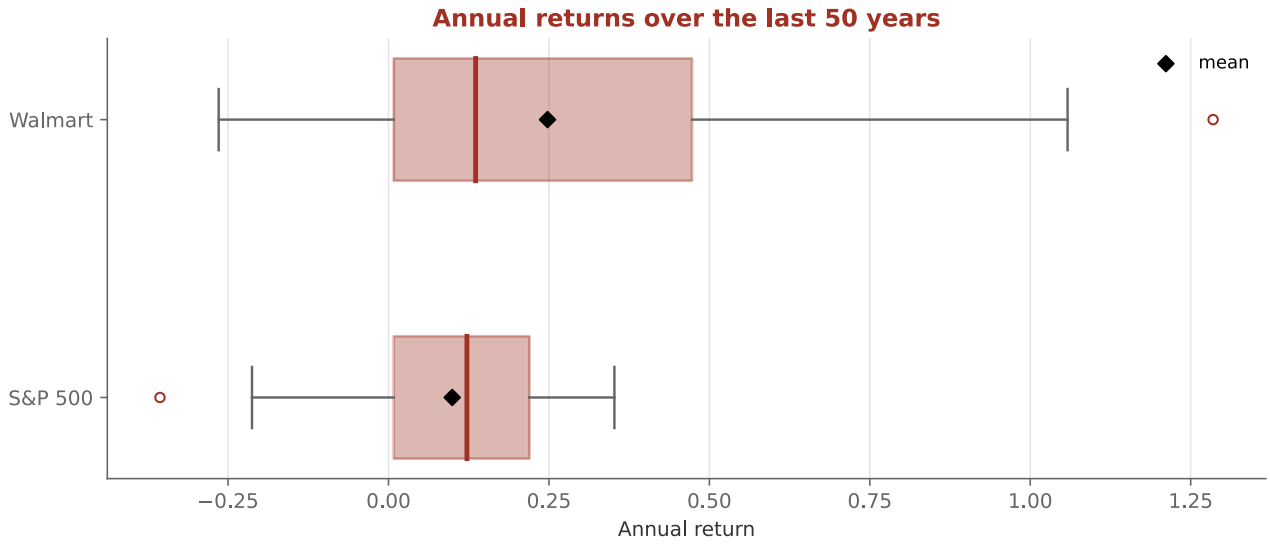
Box-and-whisker plots

The box runs from Q_1 to Q_3 with the median marked inside it, so the box *is* the interquartile range drawn to scale. The whiskers extend to the most extreme observations still within $Q_1 - 1.5 \cdot IQR$ and $Q_3 + 1.5 \cdot IQR$. Anything beyond those fences is plotted individually and is what this chapter means by an outlier.



The long lower whisker and the short upper one are the left skew, seen from a different angle. The diamond marks the mean; that it sits below the median line is the same fact the histogram showed.

Box plots earn their keep when comparing groups:



The five-number summary

By convention, five statistics describe a data set and the shape of its distribution: the **minimum**, the **first quartile**, the **median**, the **third quartile**, and the **maximum**.

	Graduate GPA	S&P 500 return	Walmart return	Unemployment rate
min	1.350	−0.3561	−0.2646	2.5
Q1	3.425	0.0081	0.0083	4.3
median	3.712	0.1222	0.1358	5.5
Q3	3.900	0.2194	0.4730	6.7
max	4.000	0.3520	1.2850	14.8

That is the box plot as a table, and it is the block Excel produces from **Data → Data Analysis → Descriptive Statistics**.

Measures of Association Between Two Variables

The **sample covariance** measures the *direction* of the linear relationship between two variables:

$$\hat{\sigma}_{xy} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}).$$

The **sample correlation coefficient** measures both the strength and the direction:

$$\hat{\rho}_{xy} = \frac{\hat{\sigma}_{xy}}{\hat{\sigma}_x \hat{\sigma}_y},$$

where $\hat{\sigma}_x$ and $\hat{\sigma}_y$ are the sample standard deviations of x and y. It always lies between -1 and $+1$.



	S&P 500 vs Walmart
n_pairs	50.000000
sd_sp500	0.156811
sd_wmt	0.344288
covariance	0.023689
correlation	0.438786

The covariance is 0.023689. Its sign says the two move together. Its magnitude says nothing you can act on, because it carries the units of both variables multiplied together — rescale returns from decimals to percentages and the covariance changes by a factor of 10,000 while the relationship does not change at all.

Dividing by both standard deviations removes the units and leaves 0.4388 — a moderate positive relationship, on a scale that runs from -1 to $+1$ and means the same thing whatever the variables were measured in.

	HPI	UR	POPN	gPOP
HPI	1.000	0.101	0.182	0.300
UR	0.101	1.000	0.388	−0.003
POPN	0.182	0.388	1.000	0.088
gPOP	0.300	−0.003	0.088	1.000

A correlation matrix is the fastest way to mislead yourself in this entire course. Four things it does not tell you:

1. **Causation.** Neither direction, nor the possibility that a third variable drives both.

2. **Non-linear relationships.** A perfect U-shape can produce a correlation of zero. Draw the scatter plot before you trust the coefficient.
3. **What outliers did to it.** A single extreme pair can create a correlation or destroy one, and the number gives no hint.
4. **Whether the sample is large enough to mean anything.** With 50 observations, a coefficient of 0.25 is not distinguishable from zero — which is a Chapter 9 question, and one we will come back to.

Laboratory 3

There are two ways to get these files to Claude. **Start with Path 1.** It works on every Claude account, including the free tier, and needs nothing turned on.

Path 1 — upload the files

Download the Laboratory 3 pack — one file. Unzip it and you will have a folder holding 11 files. Drag all 11 into a new Claude conversation, wait for each to show as attached, then paste this:

PROMPT

I have uploaded 11 files for a graduate business statistics course. Read `COURSE_CONTEXT.md` first — it explains how this course expects analytical work to be done.

Then run `03_descriptive_statistics.py` and show me its actual output, verbatim. Walk me through what each section does and what the results mean. Do not report any number you did not compute, and if anything fails, show me the error rather than describing what the output would have been.

Path 2 — let Claude fetch the files

If web access is working on your account, Claude can retrieve the files itself:

PROMPT

Open https://crooker.faculty.unlv.edu/mba775/fall2026/upload_packs/ — it is a plain page listing the files for each laboratory. Follow the links for Laboratory 3 and read `COURSE_CONTEXT.md` first; it explains how this course expects analytical work to be done.

Then run `03_descriptive_statistics.py` and show me its actual output, verbatim. Walk me through what each section does and what the results mean. Do not report any number you did not compute, and if anything fails, show me the error rather than describing what the output would have been.

If you cannot open that page, tell me plainly and I will upload the files instead.

If Claude answers that it cannot reach GitHub, **nothing is wrong with your work or with the repository.** Claude's web access is a setting, and even when it is on, Claude cannot always retrieve a file by direct link. Use Path 1 instead — it is the ordinary route, not a consolation prize.

Working in Excel

PROMPT

> Using `data/annual_returns.csv` from https://github.com/jcrooker/mba775, build an Excel workbook with two sheets. Sheet 1 holds the raw annual returns for the S&P 500 and Walmart. Sheet 2 computes, for each series, the mean, median, sample standard deviation, minimum, maximum, range, and coefficient of variation, plus the correlation between the two series — every cell a real Excel formula referencing Sheet 1, not a pasted value. Add a column chart of both return series by year. Give me the file to download.

When it opens, change one year's return to something extreme and watch which statistics move and which do not. The ones that barely move are the resistant ones, and now you know why.